CHETHAN MYSURU RADHAKRISHNA

M.S., in Data and Knowledge Engineering

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EXPERIENCE

Big Data Analyst in Project Management Metrics
Bosch Engineering GmbH
Apr 2021 – Dec 2022 Abstatt, Germany

- Worked on the analysis of data originating from software development life cycle using SPLUNK
- Clean-up and modelling of data origination from various sources including sprint planning and requirement management tools like ALM and DOORS
- Developed dashboards for the visualization and dynamic analysis of the full V-Model pertaining to requirement management
- Rolled out the generic real-time dashboards to various projects using CICD tools including bitbucket and Jenkins

Technologies: SPLUNK, Python, SQL, Jenkins, bitbucket

Software Developer
Metratec Gmbh
Jun 2019 – Feb 2021 Magdeburg, Germany

- Worked as a full stack web developer and handled API integration with react-redux dashboards
- Worked on real time data using web sockets
- Worked on event sourcing and reSolve(a CQRS framework built using node.js)
- Managed Gitlab pipeline and docker configurations with yaml

Technologies: Reactjs, nodejs, reSolve, web sockets, node-red, kafka

Software Engineer
Happiest Minds Technologies Pvt Ltd
Oct 2016 – Jan 2019 Bengaluru, India

- Contributed as a full stack web developer and worked on ASP.NET MVC applications and Web API services built on .NET framework 4.6.2 and .NET core
- Worked on react.js and redux to implement stateful front-end UI
- Worked exclusively on Health Care domain with the knowledge of online data exchange standards such as HL7
- Worked on AWS EC2, S3 buckets, Simple Mail Service and Lambda(for deploying micro services)
- As a billable resource, interacted with clients to understand requirements and effectively communicated the innovative ideas to solve their problems

Technologies: ASP.NET MVC/WebAPI, C#, Reactjs, nodejs, AWS, Jenkins, Octopus

LIFE PHILOSOPHY

"What we know is a drop and what we don't know is an ocean."

SKILLS

Python
R
C#, ASP.NET
React.js
Node.js
Java
C++
SQL
Postgres
mongoDB
SPLUNK
Kafka

LIBRARIES AND CICD

PyTorch Tensorflow Keras

scikit-learn scikit-image NLTK spaCy gensim node-red reSolve

Bitbucket Git Lab Jenkins Docker Singularity

LANGUAGES

English
German

EDUCATION

M.S. in Data and Knowledge Engineering
Otto von Guericke University
Apr 2019 - Mar 2024 Magdeburg, Germany

B.E. in Computer Science and Engineering
Vidyavardhaka College of Engineering
Jun 2012 - Jul 2016 Mysuru, India

PROJECTS

Master's Thesis: Exploration of Deep Learning based Vessel Segmentation with Less to No Supervision

Code

July 2023 – March 2024

- The study aims at learning good quality vessel segmentation using 3D patch based unsupervised learning by estimating k-class probability distribution using Attention WNet architecture.
- The research explores the possibility of generating quality vessel segmentation in the absence of manual expert annotations.
- The study also extends the 2D MIP based pipeline to incorporate the weakly annotated MIPs of the 3D volume instead of the full-volume annotations.
- The study compares the segmentation results of unsupervised, weakly-supervised and semi-supervised approaches and provides a novel performance ranking based on vessel diameters.

Technologies Used: Python, PyTorch, nibabel, scikit-image, cuda, WNet, Deep CNN

Enhancing Vessel Continuity in Deep Learning based Segmentation using Maximum Intensity Projection as Loss

Code **Paper**

April 2023 – Sep 2024

- Study focuses on challenges pertaining to segmentation of biomedical images by extending *"DS6: Deformation-aware Semi-supervised learning"* published by Dr.Chatterjee et.al., with Maximum Intensity Projection (MIP) comparisons
- Proposed Vessel Segmentation approach was validated using limited annotated samples of high resolution 7Tesla 3D-MRA volumes
- Qualitative analysis of ROI's revealed clear improvement in the continuity of vessel structures perceived by the Deep Learning network after incorporating MIP comparisons as one of the loss terms during training

Technologies Used: Python, PyTorch, nibabel, scikit-image, cuda, docker

Deep Learning Pipeline for Analysis of the 3D Morphology of the Cerebral Small Perforating Arteries

Code **Paper**

Oct 2022 – Jun 2023

- Research aims at exploring possible benefits of Unsupervised Deep Learning (DL) techniques to reduce the dependency on prior annotations in 3D Image Segmentation
- Research tasks include comparison of quality of segmentations generated using UNet and WNet based architectures with and without limited/weak annotations
- Study extends *Attention WNet* architecture with the similarity and consistency losses proposed in the *Differentiable Feature Clustering (DFC)*
- Approach is validated by performing Pathology Detection(Tumor Segmentation) using 3D MRI volumes obtained using 3Tesla and 7Tesla scanners

Technologies Used: Python, PyTorch, Docker, Singularity, scikit-image, cuda, WNet, Deep CNN

PUBLICATIONS

Journal Articles

- R. Li, S. Chatterjee, Y. Jiaerken, *et al.*, "Lumen–a deep learning pipeline for analysis of the 3d morphology of the cerebral lenticulostriate arteries from time-of-flight 7t mri," *NeuroImage*, vol. 318, p. 121 377, 2025, ISSN: 1053-8119. DOI: https://doi.org/10.1016/j.neuroimage.2025.121377.
- S. Chatterjee, H. Mattern, M. Dörner, *et al.*, "Smile-uhura challenge – small vessel segmentation at mesoscopic scale from ultra-high resolution 7t magnetic resonance angiograms," 2024. arXiv: 2411.09593 `[eess.IV]`. [Online]. Available: https://arxiv.org/abs/2411.09593.
- R. Li, S. Chatterjee, Y. Jiaerken, *et al.*, "A deep learning pipeline for analysis of the 3d morphology of the cerebral small perforating arteries from time-of-flight 7 tesla mri," *medRxiv*, 2024. DOI: 10.1101/2024.10.03.24314845. eprint: https://www.medrxiv.org/content/early/2024/10/04/2024.10.03.24314845.full.pdf.
- C. Radhakrishna, K. V. Chintalapati, S. C. H. R. Kumar, *et al.*, "Spockmip: Segmentation of vessels in mras with enhanced continuity using maximum intensity projection as loss," 2024. arXiv: 2407.08655 `[eess.IV]`. [Online]. Available: https://arxiv.org/abs/2407.08655.
- C. Radhakrishna, A. M. Vijaykumar, K. N. M. Gurkar, M. S. Akarsh, and P. K. N. Kumar, "Gesture recognition for interactive systems using kinect v2," 2016. DOI: 10.15680/IJIRCCE.2015. eprint: http://ijircce.com/admin/main/storage/app/pdf/lmTe2rGj2atzpnIAlSAWXv8svf8kR7qlAwFWd7nc.pdf.

Conference Proceedings

- S. Chatterjee, C. Radhakrishna, S. C. H. Kumar, *et al.*, "Enhancing vessel continuity in deep learning based segmentation using maximum intensity projection as loss," in *Proceedings of the ISMRM & ISMRT Annual Meeting & Exhibition*, Toronto, Canada, 2023.